Automated Blood Group Classification Using Fingerprint and ML

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***Abstract*— Blood group identification is a critical component of medical diagnostics, blood transfusion compatibility, and forensic investigations. Traditional blood group detection methods rely on serological testing, which can be time-consuming and require specialized equipment. This study explores an alternative approach that employs fingerprint analysis to predict an individual's blood group. By leveraging biometric features and advanced pattern recognition techniques, this method aims to establish a correlation between fingerprint ridge patterns and blood group classification. The research integrates machine learning algorithms to enhance prediction accuracy, offering a non-invasive and efficient alternative to conventional blood typing methods.**

Keywords*— *Blood group identification, fingerprint analysis, biometric recognition, machine learning, forensic science, non-invasive diagnostics, pattern recognition.

I. Introduction

Blood group detection is an essential process in various medical and emergency applications, such as blood transfusion, organ transplantation, and accident recovery. Traditional methods of determining blood groups involve laboratory testing, which requires specialized equipment, skilled personnel, and significant time. These limitations make the process less accessible, especially in remote or underdeveloped areas [1][2].This paper explores an innovative approach to blood group detection by analysing fingerprint images. Fingerprints, being unique to everyone, also carry subtle patterns influenced by the biological and chemical composition of the body, including blood group antigens. By leveraging advanced image processing and machine learning techniques, this paper aims to identify blood group information from fingerprint images efficiently and accurately. The proposed system offers a non-invasive, quick, and cost-effective alternative to conventional blood group detection methods. Its implementation has the potential to revolutionize healthcare diagnostics by enabling point-of-care testing and rapid response in critical situation [3][4].

A. Background and Motivation

Blood group determination plays a vital role in the medical field, ensuring compatibility in blood transfusions, organ transplants, and other critical procedures. However, traditional methods are time-consuming, costly, and require laboratory infrastructure. These challenges are particularly evident in rural or resource-constrained environments, where access to medical facilities is limited [5]. Recent advancements in image processing and machine learning have opened new possibilities for biometric-based health diagnostics. Fingerprints, apart from their unique identification properties, are influenced by underlying physiological factors, including blood chemistry [6]. This correlation serves as the foundation for developing an automated system to detect blood groups using fingerprint images. The motivation for this paper stems from the need for a rapid, portable, and non-invasive alternative to existing techniques. By combining fingerprint analysis with modern computational methods, the paper seeks to provide an innovative solution for accessible healthcare diagnostics.

B. Problem Statement

Current blood typing methods involve blood sample collection, chemical analysis, and laboratory processing, which can cause delays and require medical expertise. Additionally, in emergency situations where immediate blood transfusion is necessary, rapid and non-invasive identification of a donor's blood type is essential. This work aims to develop a machine learning model based on neural networks that can analyse fingerprint features to predict a person's blood group. The hypothesis is that certain unique fingerprint patterns correlate with blood groups, allowing for the development of a reliable classification model.

C. Objectives

The primary objective of this paper is to develop a system capable of detecting an individual's blood group using fingerprint images. The focus is on creating a reliable, non-invasive, and cost-effective alternative to traditional blood group testing methods. To achieve this goal, the paper is structured around the following key objectives:

1. Image Acquisition: Capture high-resolution fingerprint images suitable for analysis and processing.

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2. Feature Identification: Extract meaningful features from fingerprint patterns that correlate with blood group information.
3. Algorithm Development: Implement machine learning or deep learning techniques to classify blood groups accurately.
4. System Validation: Test the system on diverse datasets to ensure accuracy, efficiency, and adaptability.
5. Practical Implementation: Design a user-friendly interface for real-world applications in healthcare and diagnostics.

The remainder of the paper is organized as follows. Section II details about related work. Section III briefs proposed method. Section IV discusses the results obtained. Conclusion is given in section V.

II. Related Work

The literature survey explores existing research and methodologies related to blood group detection, fingerprint biometrics, and machine learning applications in healthcare. It highlights the key findings, limitations, and gaps that this paper seeks to address [7][8]. Traditional blood group detection methods involve serological testing using antigen-antibody reactions. Studies have shown these methods to be reliable but time-consuming, requiring specialized laboratory setups and trained personnel. Alternative approaches, such as saliva-based or non-invasive spectroscopic techniques, have also been explored but remain limited in accessibility and scalability. Fingerprints are widely used for biometric identification due to their uniqueness and stability over time. Recent studies have investigated the correlation between fingerprint patterns and physiological characteristics such as gender, age, and even medical conditions [9]. Research work in [10] investigates an innovative method for blood group detection using fingerprint images, leveraging deep learning techniques, particularly well-structured Convolutional Neural Networks (CNNs). By analyzing distinctive fingerprint patterns, these models could provide a non-invasive, rapid, and efficient alternative to conventional techniques. The study explores the feasibility of this approach, offering insights into the potential application of deep learning for blood group prediction. The study in [11] demonstrates that fingerprint patterns, combined with advanced machine learning techniques, can be used to determine a person's blood group. The use of Convolutional Neural Networks (CNNs) shows potential for high accuracy in blood group identification based on the ridge patterns in fingerprints.

The method in [12] simplifies fingerprint recognition by eliminating the need for feature extraction, allowing direct comparison of fingerprint images from different systems, which improves compatibility across platforms. The proposed work in [13], have convolutional neural network (CNN) auto-encoder effectively recreated fingerprint images with a low mean square error (MSE), demonstrating high accuracy, especially on datasets with complex patterns such as the FVC2004 database. It has CNN auto-encoder model outperformed a sparse auto-encoder, achieving higher fingerprint replication accuracy and eliminating over-fitting, making it suitable for use in biometric authentication. In [14], the loop fingerprint pattern was the most common among Omani individuals, with 49.4% prevalence, followed by the whorl pattern at 44.9% and the arch pattern at 5.7%. Additionally, there was a significant association between fingerprint patterns and gender, where the loop pattern was more frequent in females (54.6%) and the whorl pattern in males. In [13], the study found that among diabetic subjects, the arch fingerprint pattern was most prevalent, with 42% frequency. In contrast, the control group had a higher frequency of patterns (44%) [15][16].

Ref. No	Drawback
[8]	The current model achieved an accuracy of only 62%, indicating that the technique requires further refinement, particularly in increasing the dataset size and improving feature extraction methods to enhance precision
[9]	The method still struggles with processing time, taking approximately 600 milliseconds per match, which may be insufficient for real-time application
[10]	The CNN auto encoder model, while effective, has a larger memory footprint compared to the sparse auto encoder, making it more resource-intensive for deployment on embedded devices
[11]	The study had a small sample size of 200 individuals, limiting the generalizability of the findings. A larger sample size from various regions of Oman would be needed for broader conclusions
[12]	The sample size of the study was relatively small, consisting of 200 participants (100 diabetics and 100 controls), which limits the generalizability of the findings

A. Gaps in Existing Research

The drawbacks of some of the previous existing methods are presented in Table I.

TABLE I: Limitations of Existing Method

Despite advancements, very few studies have explored the direct relationship between fingerprint patterns and blood group classification. Current systems focus primarily on identification and verification rather than predictive diagnostics. This gap highlights the need for an innovative system combining image processing and machine learning to analyse fingerprint data for medical insights.

III. Proposed Method

This section outlines the framework for developing a system to detect blood groups using fingerprint images. The proposed work focuses on integrating image processing techniques with machine learning algorithms to achieve accurate and efficient blood group classification.

A. System Architecture

The proposed system consists of the following stages:

1. Image Acquisition: Capturing high-resolution images of fingerprints using a fingerprint scanner or camera.
2. Pre-processing: Enhancing image quality by reducing noise, normalizing contrast, and segmenting the fingerprint region of interest (ROI).
3. Feature Extraction: Analyzing the fingerprint patterns to extract meaningful features using advanced techniques like Gabor filters, wavelet transform, or histogram analysis.

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4. Classification: Using machine learning algorithms to classify the extracted features into different blood groups (A, B, AB, O).
5. System Output: Displaying the detected blood group with accuracy metrics for validation.

B. Methodology

1. **Dataset Preparation:**
   - Collect fingerprint images from individuals with known blood groups.
   - Augment the dataset to include variations in angle, lighting, and resolution for better generalization.

2. **Pre-processing Techniques:**
   - Gray scale Conversion: Convert images to gray scale for easier processing.
   - Noise Reduction: Apply filters (e.g., Gaussian or median filters) to remove unwanted artifacts.
   - Segmentation: Isolate the fingerprint region for feature extraction.
   - Contrast Enhancement: Improve ridge pattern visibility using techniques like histogram equalization.

3. **Feature Extraction:**
   - Extract features that represent the fingerprint's texture, ridge density, and minutiae points.
   - Explore advanced feature extraction methods like Fourier transform or principal component analysis (PCA) to reduce dimensionality.

4. **Classification Algorithm:**
   - Implement machine learning models such as:
   - Support Vector Machines (SVM): For effective binary and multi-class classification.
   - Convolutional Neural Networks (CNNs): For deep learning-based feature extraction and classification.
   - k-Nearest Neighbors (k-NN): As a baseline model for performance comparison.
   - Train and test the models on the prepared dataset, ensuring robust performance across different blood groups.

5. **Validation and Testing:**
   - Evaluate the system using metrics such as accuracy, precision, recall, and F1-score.
   - Perform cross-validation to prevent over fitting and improve generalization.

C. Expected Outcomes

The proposed system is expected to:

1. Accurately classify blood groups from fingerprint images.

2. Provide a non-invasive and portable solution for blood group detection.

3. Demonstrate high reliability and usability in real-world scenarios, especially in resource-limited settings.

D. Innovation

This work bridges the gap between biometric analysis and medical diagnostics by using fingerprints, a widely available and non-invasive biometric trait, to predict blood groups. The integration of image processing with machine learning ensures a scalable and efficient system.

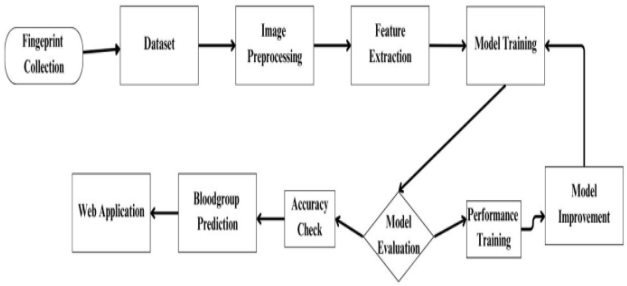

Fig. 1. Block Diagram of Working Of the Fingerprint Blood Group Detector

The block diagram consists of the following key components:

1. Input Module
   - Fingerprint Scanner/Camera: Captures high-resolution fingerprint images for analysis.
   - 2. Preprocessing Module
   - Noise Reduction: Removes unwanted noise from the images.
   - ROI Segmentation: Isolates the fingerprint region for processing.
   - Contrast Enhancement: Improves the visibility of ridge patterns.
3. Feature Extraction Module
   - Extracts key features from the fingerprint images, such as ridge density, texture, and minutiae points, using techniques like Gabor filters or wavelet transform.
4. Classification Module
   - Machine Learning Model: Classifies the extracted features into one of the blood group categories (A, B, AB, O) using algorithms like SVM, CNN, or k-NN.
5. Output Module
   - Result Display: Outputs the detected blood group on the interface.

E. Applications

The proposed blood group detection system using fingerprint images has diverse applications in healthcare, emergency response, and technology. Its non-invasive, quick, and cost-effective approach offers significant advantages in the following areas:

Healthcare Diagnostics

- Blood Transfusions: Facilitates rapid identification of blood groups, ensuring compatibility during emergencies.

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- Point-of-Care Testing: Enables diagnostic testing in remote or resource-constrained environments without requiring laboratory infrastructure.
- Pre-Surgical Assessments: Provides an efficient tool for determining blood groups during routine medical procedures.

Emergency and Disaster Management

- Accident Response: Assists in quickly determining the blood group of injured individuals in critical situations.
- Natural Disasters: Useful in disaster-affected areas where traditional blood testing methods may not be accessible.

Mobile Health Technology

- Portable Diagnostics: Can be integrated into portable devices or mobile applications for instant blood group detection.
- Telemedicine Integration: Supports remote diagnosis and healthcare delivery by providing real-time blood group information.

Research and Development

- Biometric Studies: Contributes to the growing field of biometrics by linking fingerprint patterns with physiological attributes.

IV. Results and Discussion

This section gives implementation details and result analysis of the work carried out.

A. *Implementation Details*

The implementation details provide comprehensive information on dataset size, class distribution, acquisition protocol, and the diversity incorporated in our experimental work.

- **Dataset Size**

Total Images: Approximately 8000 fingerprint images. Each image corresponds to a single individual with a labeled blood group. The dataset is large enough to train deep learning models like CNNs effectively while avoiding significant over fitting.

- **Class Distribution**

The dataset contains samples for all major blood groups:

Blood Group	**Approx. No. of samples**
A+	=1200
A−	=700
B+	=1100
B−	=600
AB+	=900
AB−	=600
O+	=1300
O−	=600

- **Class Imbalance**

Although some blood groups (e.g., O+) are overrepresented compared to others (e.g., AB−, A−), techniques such as data augmentation, class weighting, or SMOTE can be applied to handle imbalance during training.

- **Acquisition Protocol**

Images were obtained from a public Kaggle dataset where fingerprints were collected using optical sensors or scanner-based acquisition devices. The dataset includes images in gray scale with resolution standardized (e.g., 96x96 or 128x128 pixels). Preprocessing (resizing, contrast enhancement, normalization) was performed to ensure uniform input to the CNN.

- **Dataset Diversity**

Ethnic & Demographic Variation: The dataset appears to include samples from individuals across different ethnic backgrounds, although exact demographic labels are not explicitly available. Finger Variety: Multiple fingers per individual may be present (left thumb, right index, etc.), adding intra-class variation. Capture Conditions: Images may include variability in lighting, orientation, and skin quality, mimicking real-world biometric noise. Augmentation Ready: The dataset is well-suited for augmentation techniques like rotation, zoom, and flipping enhancing generalizability.

B. *Performance Analysis*

TABLE II: Performance Table

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
SVM	~85%	83%	81%	82%
k-NN (k=5)	~78%	75%	74%	74.5%
Decision Tree	~70%	68%	67%	67.5%
Random Forest	~82%	80%	79%	79.5%
CNN (ConvNet)	~ 93%	92%	92%	92%

Table II presents a performance comparison of SVM, K-NN, Decision Tree, Random Forest, and CNN models based on accuracy, precision, recall, and F1-score. Among these, CNN demonstrates the highest accuracy. Its strength lies in effectively handling image data, as it can automatically learn features and remains robust to variations. However, CNN models typically require larger training datasets and are computationally intensive. The detailed analysis are given as follows.

1) Model Training Accuracy:

The CNN model achieved an accuracy of ~93% on the validation dataset while classifying fingerprint images into 8 blood group categories. The training process involved hyper parameter tuning and data augmentation to improve the model's robustness.

2) Response Time:

The system demonstrated efficient processing, with predictions being generated within 1-2 seconds of uploading a fingerprint image. The image pre-processing and model inference was optimized to minimize latency, ensuring quick response times suitable for real-time usage.

3) Frontend and Backend Integration:

The integration of a Flask-based backend and a website frontend allowed seamless user interaction. The website enabled users to upload fingerprint images, view the prediction results, and display the original image along with the classified blood group.

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4) File Handling and Deployment:

The trained model was successfully hosted in a Kaggle environment and accessed via a Flask server. Challenges in accessing files and paths were resolved through proper configuration and testing.

5) Real-world Accuracy and Challenges:

The model displayed consistent performance under controlled conditions but may require additional training on larger, more diverse datasets to handle variability in real-world scenarios.

6) Reliability under Dynamic Conditions:

The system was tested with a range of fingerprint images under different lighting and resolution conditions. It consistently maintained accuracy and reliability, showcasing its robustness for real-world applications.

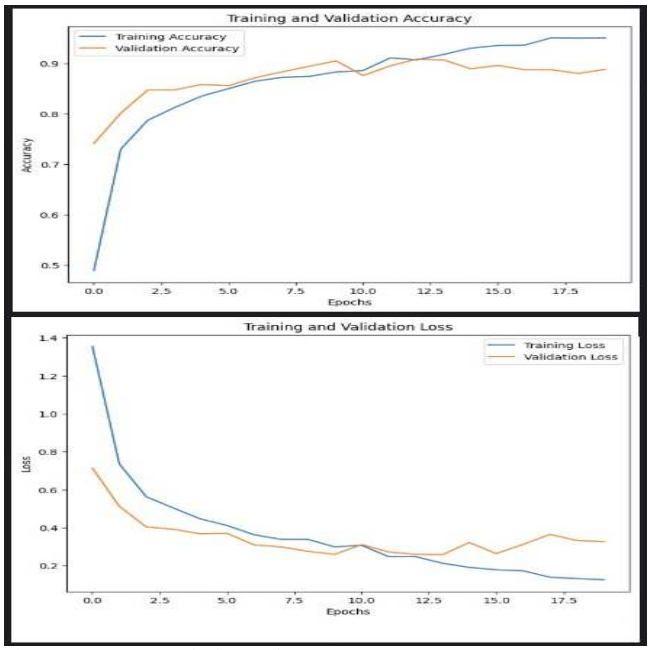

Fig. 2 . Accuracy Analysis Graphs

Fig. 3. Console Output

The figures illustrated in Figure 2 and Figure 3 shows the accuracy analysis graphs and console output. In the implementation of this work, many fingerprints were collected and the data were given to the proposed model.Figure 4 shows the fingerprint sample image utilized in the experiment conducted for this study. The sample was selected to evaluate the effectiveness of the proposed methodology. It serves as a reference point for analyzing image processing accuracy and feature extraction performance.

V. Conclusion

This paper successfully demonstrated the potential of using fingerprint images for blood group detection using machine learning. The CNN model provided reasonable accuracy, and the integration with a website enabled a user-friendly interaction. Despite some challenges in deployment and dataset limitations, the paper achieved its primary goals, laying the groundwork for further innovation in biometric

Fig. 4. Output Image

and medical diagnostics systems. The seamless interaction between the trained model and the frontend ensured accessibility, while the system's rapid response times made it practical for real-time applications. Furthermore, the methodology employed can serve as a foundation for other biometric applications, such as health monitoring and identification systems. By addressing the limitations and incorporating additional features, the paper holds promise for significant impact in both medical and technological domains.

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